

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

TixelERP Enterprise Resource Planning System

Prepared For: TixelTech Private Limited

CIN: U62013OD2025PTC051591

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Document Conventions

This document has been prepared in conformance with ISO/IEC/IEEE 29148:2018 (Systems and Software Engineering — Life Cycle Processes — Requirements Engineering). The following conventions are observed throughout this document:

Shall: Indicates a mandatory requirement that the system must satisfy.

Should: Indicates a recommended requirement that is expected to be fulfilled unless a justified reason exists for deviation.

May: Indicates an optional requirement or feature that may be implemented at the discretion of the development team, subject to mutual agreement.

Critical / High / Medium / Low: Priority classifications assigned to individual features as defined in the governing Agreement (Annexure A).

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1. INTRODUCTION

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a comprehensive, detailed, and unambiguous description of the functional and non-functional requirements pertaining to the Enterprise Resource Planning (ERP) system, hereinafter referred to as “TixelERP,” to be designed, developed, tested, deployed, and maintained by Trackepay Fintech (OPC) Private Limited (“Trackepay” / “Developer”) for and on behalf of TixelTech Private Limited (“TixelTech” / “Client”).

This SRS constitutes a formal deliverable under Milestone M1 (Kickoff & Planning) of the Master Technology Development, Intellectual Property Ownership & Revenue Sharing Agreement dated 2nd March 2026 (“Governing Agreement”) executed between the Parties. Upon written acceptance of this SRS by TixelTech, the Milestone M1 deliverable obligations of Trackepay, insofar as they relate to the Software Requirements Specification, shall be deemed fulfilled.

This document serves as the authoritative reference for all stakeholders, including project sponsors, project managers, software architects, developers, quality assurance teams, and system administrators, and shall form the basis for system design, development, testing, acceptance, and validation activities throughout the project lifecycle.

1.2 Project Background

TixelTech Private Limited, a company incorporated under the Companies Act, 2013 (CIN: U62013OD2025PTC051591), is engaged in conceptualising, owning, branding, and commercialising a proprietary Enterprise Resource Planning (ERP) Software Platform designed to serve multiple industries including, but not limited to, manufacturing, services, hospitality, construction, and general enterprises.

Trackepay Fintech (OPC) Private Limited (CIN: U74999OR2022OPC040536) possesses technical expertise in software architecture, full-stack application development, cloud infrastructure management, and fintech-grade systems development. Pursuant to the Governing Agreement, TixelTech has appointed Trackepay as its exclusive technology development vendor for the design, development, deployment, maintenance, and enhancement of TixelERP.

The total consideration for the initial development of TixelERP, covering all thirteen (13) core modules as specified herein, is fixed at ₹8,00,000/- (Rupees Eight Lakhs Only), payable in accordance with the milestone-based payment schedule defined in Annexure F of the Governing Agreement.

1.3 Scope of the System

TixelERP is envisioned as a comprehensive, multi-tenant, cloud-hosted, Software-as-a-Service (SaaS) Enterprise Resource Planning platform. The system shall integrate and automate the following core business functions through thirteen (13) interconnected modules:

Module	Name
Module 1	Login/Signup & Authentication System
Module 2	Home Screen & User Interface
Module 3	Dashboard & Data Visualisation
Module 4	Organisation Management
Module 5	Sales & CRM
Module 6	Marketing Automation
Module 7	Supply Chain Management (SCM)
Module 8	Human Resource Management (HRM)
Module 9	Finance & Accounting
Module 10	Task/Services/Project Management
Module 11	Website Builder & Content Management
Module 12	Report Generator
Module 13	Office/Workspace Collaboration Tools

In addition to the above, the system shall incorporate cross-cutting platform features including multi-tenant architecture, mobile applications (iOS and Android), offline mode, advanced search, data export/import, activity logging, a unified notifications engine, RESTful API access, webhooks, and custom field support.

The scope further encompasses web application development (responsive, cross-browser), native or hybrid mobile application development, backend API development, database design and implementation, cloud infrastructure setup, CI/CD pipeline configuration, comprehensive testing (unit, integration, functional, performance, security), and complete technical, user, and operational documentation.

1.4 Relationship to the Governing Agreement

This SRS document is subordinate to and governed by the Master Technology Development, Intellectual Property Ownership & Revenue Sharing Agreement dated 2nd March 2026. In the event of any conflict or inconsistency between this SRS and the Governing Agreement, the provisions of the Governing Agreement shall prevail unless expressly stated otherwise in a signed addendum.

All intellectual property rights in and to TixelERP, including all components, derivatives, and related materials described herein, shall vest solely and exclusively with TixelTech from the moment of creation, in accordance with Clause 7 of the Governing Agreement.

1.5 Intended Audience

Audience	Role	Relevant Sections
Project Sponsors / Business Stakeholders	Approval of scope, features, and business alignment	Sections 1, 2, 4, 11, 12

Audience	Role	Relevant Sections
Project Managers	Planning, tracking, resource allocation, milestone management	Sections 1, 2, 4, 10, 11, 12
Software Architects & Developers	System design, development, and technical implementation	Sections 3, 4, 5, 6, 7, 8, 9
Quality Assurance & Testing Team	Test planning, test case design, validation, and verification	Sections 4, 5, 7, 10
System Administrators & DevOps	Deployment, infrastructure management, monitoring	Sections 3, 8, 9
Legal / Compliance	Contractual alignment, regulatory compliance	Sections 1, 2, 7, 12, 13

1.6 Definitions, Acronyms, and Abbreviations

Term / Acronym	Definition
ERP	Enterprise Resource Planning — an integrated software system for managing core business processes.
SRS	Software Requirements Specification — a formal document describing functional and non-functional system requirements.
SaaS	Software as a Service — a cloud-based delivery model where software is hosted centrally and accessed via the internet.
Multi-Tenant	A software architecture where a single instance of the application serves multiple tenant organisations with logical data isolation.
API	Application Programming Interface — a set of protocols enabling communication between software systems.
REST	Representational State Transfer — an architectural style for designing networked applications.
UI	User Interface — the visual layer through which users interact with the system.
UX	User Experience — the overall experience of a user when interacting with the system.
RBAC	Role-Based Access Control — a method of restricting system access based on user roles.
JWT	JSON Web Token — a compact, URL-safe means of representing claims to be transferred between two parties.
OAuth 2.0	An open standard for access delegation, commonly used for token-based authentication.
CI/CD	Continuous Integration / Continuous Deployment — automated build, test, and deployment pipelines.
HRM	Human Resource Management.
CRM	Customer Relationship Management.
SCM	Supply Chain Management.
POS	Point of Sale.

Term / Acronym	Definition
GST	Goods and Services Tax (India).
DPDP Act	Digital Personal Data Protection Act, 2023 (India).
WCAG	Web Content Accessibility Guidelines.
OWASP	Open Web Application Security Project.
SLA	Service Level Agreement.
UAT	User Acceptance Testing.
MVP	Minimum Viable Product.
PLM	Product Lifecycle Management.
BOM	Bill of Materials.
CDN	Content Delivery Network.
TLS	Transport Layer Security.
AES	Advanced Encryption Standard.
VoIP	Voice over Internet Protocol.
Governing Agreement	The Master Technology Development, Intellectual Property Ownership & Revenue Sharing Agreement dated 2nd March 2026 between TixelTech and Trackepay.

1.7 System Overview

TixelERP is a comprehensive, cloud-hosted, multi-tenant Enterprise Resource Planning platform designed to integrate and automate the full spectrum of business operations for organisations across multiple industries. The platform shall provide a unified, web-based interface supplemented by native or hybrid mobile applications for iOS and Android, enabling authorised users to access the system securely from any location and device.

The system shall comprise thirteen (13) core modules operating on a shared database with tenant-level data isolation, ensuring that data entered in one module is available in real time to other modules within the same tenant context. Role-Based Access Control (RBAC) shall govern user permissions at a granular level, ensuring that each user can access only those modules, features, and data elements that are pertinent to their assigned role and responsibilities.

TixelERP shall support key business operations including, but not limited to: user authentication and access management; organisation administration; dashboard and data visualisation; sales and customer relationship management; marketing automation; supply chain and inventory management; human resource management including recruitment, attendance, leave, and payroll; finance and accounting including invoicing, expense management, and statutory compliance; project and task management including field service and helpdesk; website building and content management; industry-specific report generation; and office/workspace collaboration tools including messaging, voice calls, and video conferencing.

The platform shall be designed with scalability, security, and performance as foundational principles, enabling seamless horizontal scaling to accommodate increasing user loads, data volumes, and transaction throughput as TixelTech's client base expands.

2. OVERALL DESCRIPTION

2.1 Product Perspective

TixelERP shall function as a centralised, cloud-hosted SaaS platform that integrates multiple organisational processes into a unified system. Rather than operating disparate standalone applications for different departments, TixelERP shall provide a single platform where all departments within a tenant organisation can access and manage their respective operations through dedicated modules while sharing a common, tenant-isolated data repository.

The system shall be architected as a multi-tenant application, wherein a single deployment of the platform shall serve multiple client organisations (tenants). Each tenant's data shall be logically isolated at the database level, ensuring complete confidentiality and separation of business data between tenants. This architecture shall enable TixelTech to onboard new clients without provisioning separate infrastructure, thereby supporting cost-effective scaling.

TixelERP shall also support integration with external systems including payment gateways, email service providers, SMS gateways, WhatsApp Business API, cloud storage services, calendar synchronisation services, social media platforms, third-party order aggregators, and accounting software (such as Tally ERP), through well-defined RESTful APIs and webhook mechanisms.

2.2 Product Functions (High-Level)

The major functional capabilities of TixelERP are summarised as follows:

Function Area	Description
User & Access Management	Secure authentication (OAuth 2.0, JWT, MFA), role-based access control, session management, audit logging.
Organisation Administration	Subscription management, organisational hierarchy, notice board, calendar, notes, digital signatures, contracts, form builder, knowledge base, report generator.
Dashboard & Visualisation	Interactive, real-time dashboards for projects, finance, attendance, resources, and geolocation tracking with configurable chart types.
Sales & CRM	Lead management, opportunity tracking, quotation and invoicing, point of sale (POS), subscriptions, and route planning.
Marketing Automation	Email/SMS/WhatsApp campaigns, social media management, events management, and surveys.
Supply Chain Management	Inventory management, manufacturing orders, product lifecycle management, maintenance, and quality control.
Human Resource Management	Employee database, recruitment, holidays and time off, attendance and time tracking, appraisals, and fleet management.
Finance & Accounting	Chart of accounts, journal entries, general ledger, financial statements, invoicing and payments, expense management, payroll with statutory compliance, bill settlements, and document management.
Project Management	Project creation, task management, Gantt charts, Kanban boards, timesheets, field service management, helpdesk/ticketing, planning/scheduling, and project file management.

Function Area	Description
Website Builder	Drag-and-drop website builder, blog management, forum/Q&A, and live chat.
Report Generator	Industry-specific and custom report templates with configurable parameters.
Collaboration Tools	Word processor, spreadsheet, presentation tool, email platform, internal messaging, voice calls, and video conferencing.

2.3 User Classes and Characteristics

The following user classes have been identified for TixelERP. Each user class represents a distinct category of system users with varying access levels, responsibilities, and interaction patterns:

User Class	Description	Priority
Super Administrator	TixelTech platform-level administrator with full access to tenant management, system configuration, and platform monitoring.	Critical
Tenant Administrator	Organisation-level administrator responsible for managing their organisation's settings, users, roles, permissions, and subscription.	Critical
Manager	Department or team-level manager with access to team data, approval workflows, reports, and dashboards relevant to their functional area.	High
Employee / Staff	Standard system user with access to self-service functions (attendance, leave, expenses, tasks) and module-specific features as permitted by their assigned role.	High
Sales Staff	Users engaged in sales activities including lead management, quotation creation, order processing, POS operations, and customer interaction tracking.	High
HR Personnel	Users responsible for managing employee records, recruitment, attendance, leave, payroll processing, and appraisals.	High
Finance Staff	Users responsible for invoicing, payment recording, expense management, payroll processing, and financial reporting.	High
Field Technician	Mobile-first users performing field service activities, work order execution, and on-site data capture.	Medium
External Customer	End customer of the tenant organisation accessing self-service portals for ticketing, order tracking, or communication.	Medium

2.4 Operating Environment

TixelERP shall be designed to operate within the following technical environment:

2.4.1 Client-Side Environment

Users shall access TixelERP through modern web browsers including Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari (latest two versions of each). The web interface shall be responsive, supporting desktops, laptops, and tablets. Additionally, native or hybrid mobile applications shall be provided for Android (version 8.0 and above) and iOS (version 13.0 and above).

2.4.2 Server-Side Environment

The application backend shall be developed using the React or Express.js web framework. The server-side environment shall comprise application servers running Node with WSGI/ASGI, a MongoDB relational database management system, Redis or equivalent in-memory cache/session store, a Celery-based task queue for asynchronous processing and scheduled jobs, and a reverse proxy/load balancer (Nginx or equivalent).

2.4.3 Cloud Infrastructure

TixelERP shall be hosted on a cloud platform (to be determined and approved by TixelTech). The cloud infrastructure shall support auto-scaling, load balancing, managed database services, object storage, content delivery network (CDN) integration, and robust monitoring and logging capabilities.

2.4.4 Network Environment

The system shall require stable internet connectivity for all operations. All data transmission between client and server shall be encrypted using TLS 1.2 or higher. Internal service-to-service communication within the cloud infrastructure shall also be secured.

2.5 Design and Implementation Constraints

The following constraints shall govern the design and implementation of TixelERP:

Constraint	Description
Technology Stack	The system shall be developed using Next.js, React (frontend), expressjs, React, Node (backend), and MongoDB (database). Any deviation from this approved technology stack shall require prior written approval from TixelTech, in accordance with Clause 4.4 of the Governing Agreement.
Multi-Tenancy	The system shall implement multi-tenant architecture with logical data isolation at the database level. Each tenant's data shall be completely isolated from other tenants.
Security	The system shall comply with OWASP secure coding guidelines, implement AES-256 encryption at rest, TLS 1.2+ in transit, and adhere to the Digital Personal Data Protection Act, 2023 (India).
Performance	Page load times shall not exceed 3 seconds for standard operations; API response times shall not exceed 500 milliseconds for standard queries, as specified in Clause 4.5.1(d) of the Governing Agreement.
Accessibility	The web interface shall comply with WCAG 2.1 Level AA standards.
Browser Compatibility	The web application shall be compatible with Chrome, Firefox, Safari, and Edge (latest two versions of each).
Budget	The total development cost shall not exceed ₹8,00,000/- (Rupees Eight Lakhs Only) for the initial scope, with a cost cap of ₹10,00,000/- (Rupees

Constraint	Description
	Ten Lakhs Only) inclusive of any approved change requests, as per Clause 5.5 of the Governing Agreement.
Regulatory Compliance	The system shall comply with applicable Indian data protection laws, GST regulations for invoicing, and statutory payroll requirements (PF, ESI, TDS, Professional Tax).

2.6 Assumptions and Dependencies

The following assumptions and dependencies have been identified:

2.6.1 Assumptions

- (a) TixelTech shall provide timely feedback, approvals, and decisions within the timeframes stipulated in the Governing Agreement (five (5) Business Days for deliverable review, as per Clause 4.6.4).
- (b) TixelTech shall provide all necessary business requirements, domain knowledge, sample data, and access credentials required for development and testing.
- (c) End users of TixelERP shall receive adequate training to understand system features and workflows, as facilitated by the documentation and training materials provided by Trackepay.
- (d) Users shall have reliable internet or internal network access when interacting with the system.
- (e) The data entered into the system by users shall be accurate and properly maintained.

2.6.2 Dependencies

- (a) Third-party services: The system depends on external services including payment gateways, email/SMS service providers, WhatsApp Business API, cloud hosting services, and map/geolocation APIs. Availability, pricing, and API stability of these services are outside Trackepay's control.
- (b) Cloud infrastructure: System performance and availability depend on the reliability of the chosen cloud hosting provider.
- (c) Open-source frameworks and libraries: The application relies on Next Js, React, Express, and other open-source components that must be properly maintained by their respective communities.
- (d) Mobile platform policies: App store submission and availability for iOS and Android depend on compliance with Apple App Store and Google Play Store policies.

3. SYSTEM ARCHITECTURE & TECHNOLOGY STACK

3.1 Technology Stack

Layer	Technology	Remarks
Frontend (Web)	Next.js, React	Responsive, mobile-first design; WCAG 2.1 AA compliance
Frontend (Mobile)	Flutter	Android 8.0+, iOS 13.0+; feature parity with web where feasible
Backend Framework	React, Express.js, Node	Express REST Framework for API development; WSGI/ASGI deployment
Database	MongoDB	Relational database with tenant-level data isolation
Caching	Redis / Memcached	Session management, query caching, real-time data
Task Queue	Celery + Redis/RabbitMQ	Asynchronous tasks, scheduled jobs, background processing
Authentication	OAuth 2.0, JWT, TOTP (MFA)	Secure token-based authentication per RFC 6238
Password Hashing	bcrypt or Argon2	Industry-standard password hashing algorithms
Cloud Hosting	To Be Determined (TBD)	Subject to TixelTech approval; shall support auto-scaling and managed services
Reverse Proxy / LB	Nginx or equivalent	Load balancing, SSL termination, static file serving
CI/CD	GitHub Actions / GitLab CI / Jenkins	Automated build, test, and deployment pipelines
Version Control	Git (GitHub / GitLab)	GitFlow branching strategy
Monitoring & Logging	CloudWatch / Prometheus / ELK Stack	Application and infrastructure monitoring, centralised logging
CDN	To Be Determined	Static asset delivery for performance optimisation
Visualisation	Chart.js / D3.js / ApexCharts	Interactive dashboards and reporting charts

3.2 Multi-Tenant Architecture

TixelERP shall implement a shared-database, shared-schema multi-tenant architecture with tenant identification enforced at the application layer. The key design principles are as follows:

(a) Each tenant shall be uniquely identified by a tenant identifier associated with every data record in the database. All database queries shall be scoped to the requesting tenant's context, ensuring complete logical data isolation.

(b) The Express middleware layer shall automatically inject the tenant context into every request based on the authenticated user's tenant association.

- (c) Application-level Row-Level Security (RLS) shall be enforced through a scoped database access layer (e.g., a custom Express QuerySet manager) to prevent cross-tenant data leakage.
- (d) Tenant-specific configurations, branding, and feature toggles shall be supported to allow per-tenant customisation without code changes.
- (e) The architecture shall support horizontal scaling to accommodate a growing number of tenants without degradation of performance or isolation guarantees.

3.3 High-Level System Architecture

The system shall follow a layered architecture comprising the following tiers:

Presentation Layer: Responsive web application (Next Js / React) and mobile applications communicating with the backend via RESTful APIs.

Application Layer: Express/Node-based backend handling business logic, authentication, authorisation, data validation, and orchestration of services.

Data Layer: MongoDB database with tenant-scoped data access, Redis cache for performance optimisation, and object storage for file/document management.

Integration Layer: RESTful APIs, webhooks, and adapters for third-party service integrations (payment gateways, email/SMS providers, calendar services, etc.).

Infrastructure Layer: Cloud-hosted environment with auto-scaling, load balancing, CDN, monitoring, logging, and automated backup/recovery.

3.4 API Architecture

All inter-service communication and client-server interaction shall be conducted through RESTful APIs built using Node REST Framework. The API architecture shall adhere to the following standards:

- (a) All APIs shall follow RESTful conventions with proper HTTP methods (GET, POST, PUT, PATCH, DELETE) and meaningful HTTP status codes.
- (b) API documentation shall be auto-generated and maintained using Swagger/OpenAPI specification.
- (c) API versioning shall be implemented to support backward compatibility during upgrades.
- (d) Rate limiting and throttling shall be implemented to prevent abuse and ensure fair usage.
- (e) All API endpoints shall enforce authentication and authorisation checks.

4. FUNCTIONAL REQUIREMENTS

This section defines the detailed functional requirements for all thirteen (13) core modules of TixelERP, as specified in Annexure A of the Governing Agreement. Each module is described with its sub-modules, individual features, feature identifiers, descriptions, and priority classifications. The feature identifiers used herein correspond directly to those defined in the Governing Agreement to ensure traceability.

4.1 Module 1: Login/Signup & Authentication System

The Login/Signup & Authentication System module shall be responsible for managing user registration, identity verification, session management, and access control across the TixelERP platform. This module constitutes the security foundation of the system and shall enforce strict authentication and authorisation protocols to ensure that only verified and authorised users gain access to the platform.

Technical Specifications: Authentication Protocol: OAuth 2.0 / JWT (JSON Web Tokens); Password Hashing: bcrypt or Argon2; Session Storage: Redis or equivalent in-memory store; MFA: TOTP compliant with RFC 6238.

Feature ID	Feature Name	Description	Priority
AUTH-001	User Registration	New user signup with email/mobile verification	Critical
AUTH-002	User Login	Secure login via email/username and password	Critical
AUTH-003	Password Manager	Password strength validation, reset, and recovery via email/SMS	Critical
AUTH-004	Multi-Factor Authentication (MFA)	Optional 2FA via OTP (SMS/Email/Authenticator app)	High
AUTH-005	Single Sign-On (SSO)	Integration with Google/Microsoft/LinkedIn SSO	Medium
AUTH-006	Session Management	Secure session handling, timeout, and concurrent session control	Critical
AUTH-007	Role-Based Access Control (RBAC)	User roles: Admin, Manager, Employee, and Custom roles with granular permissions	Critical
AUTH-008	Password Policy Enforcement	Configurable policies including minimum length, complexity, and expiry requirements	High
AUTH-009	Account Lockout	Auto-lock after configurable number of failed login attempts, with CAPTCHA support	High
AUTH-010	Audit Trail	Log all login/logout activities with IP address, device type, and timestamp	High

4.2 Module 2: Home Screen & User Interface

The Home Screen & User Interface module shall serve as the primary landing interface for all authenticated users. It shall provide a personalised, responsive dashboard with quick access to frequently used modules, notifications, and system-wide search capabilities.

Technical Specifications: Responsive Design: Next.js, React or equivalent framework; Mobile-First: Touch-optimised for iOS and Android; Accessibility: WCAG 2.1 Level AA compliance.

Feature ID	Feature Name	Description	Priority
HOME-001	Responsive Dashboard	Adaptive layout for web (desktop, tablet) and mobile platforms	Critical
HOME-002	Quick Access Widgets	Pinnable shortcuts to frequently used modules and features	High
HOME-003	Notifications Centre	Real-time notifications (in-app, push, email) with read/unread status	High
HOME-004	User Profile Menu	Access to profile settings, preferences, and logout	Critical
HOME-005	Global Search Bar	System-wide search across all modules (customers, projects, documents, employees)	High
HOME-006	Theme Customisation	Light/dark mode toggle and colour scheme preferences	Medium
HOME-007	Multi-Language Support	English (default), Hindi, and extensible framework for additional languages	Medium
HOME-008	Help/Support Access	Quick links to help documentation, support tickets, and knowledge base	Medium

4.3 Module 3: Dashboard & Data Visualisation

The Dashboard & Data Visualisation module shall provide interactive, real-time dashboards delivering key business metrics across projects, finance, attendance, resources, and geolocation tracking. The module shall comprise five (5) sub-modules, each addressing a specific domain of business intelligence.

Technical Specifications: Chart Types: Line, Bar, Pie, Donut, Area, Scatter, and Heatmaps; Libraries: Chart.js, D3.js, or ApexCharts; Export: PDF, Excel, and PNG export of dashboards.

Sub-Module 3A: Project/Task/Service Dashboard

Feature ID	Feature Name	Description	Priority
DASH-P-001	Time-based Views	Dashboard views: Week, Month, Year, and Total	Critical
DASH-P-002	Project Overview	Summary of active projects: scope, status, and timeline	Critical
DASH-P-003	Location-based View	Project details by location with map integration	High

Feature ID	Feature Name	Description	Priority
DASH-P-004	Resource Allocation	Human resources, assets, and vendors engaged in each project	High
DASH-P-005	Project Status Tracker	Real-time status: Not Started, In Progress, On Hold, Completed	Critical
DASH-P-006	Filter by Project	View specific project or aggregate across all projects	High

Sub-Module 3B: Finance Dashboard

Feature ID	Feature Name	Description	Priority
DASH-F-001	Revenue Analytics	Organisation-wide revenue: Week, Month, Year, Total	Critical
DASH-F-002	Project Financials	Per-project expenditure, profit/loss, and margins	Critical
DASH-F-003	Pending Recoveries	Outstanding invoices and payment due tracking	High
DASH-F-004	Expense Breakdown	Category-wise expense analysis with charts	High
DASH-F-005	Cash Flow Visualisation	Cash inflows vs outflows over time	High
DASH-F-006	Budget vs Actual	Comparison charts for budget adherence	Medium

Sub-Module 3C: Attendance Dashboard

Feature ID	Feature Name	Description	Priority
DASH-A-001	Daily Attendance	Real-time employee attendance status	Critical
DASH-A-002	Attendance Trends	Weekly/monthly attendance patterns and absenteeism analysis	High
DASH-A-003	Late/Early Departures	Track punctuality metrics across the organisation	Medium
DASH-A-004	Leave Summary	Current leave balances and pending approvals	High

Sub-Module 3D: Resources Dashboard

Feature ID	Feature Name	Description	Priority
DASH-R-001	Human Resource Utilisation	Current allocation, availability, and workload	High
DASH-R-002	Inventory Status	Stock levels, usage, and current status	High

Feature ID	Feature Name	Description	Priority
DASH-R-003	Asset Tracking	Equipment utilisation and maintenance status	Medium
DASH-R-004	Resource Forecasting	Predict future resource requirements based on trends	Low

Sub-Module 3E: Map & Tracking

Feature ID	Feature Name	Description	Priority
DASH-M-001	Live Location Tracking	Real-time GPS tracking of employees, assets, and vehicles	High
DASH-M-002	Project Site Mapping	Map view of all project locations	High
DASH-M-003	Route History	Historical movement tracking and playback	Medium
DASH-M-004	Geofencing Alerts	Alerts when resources enter/exit defined zones	Medium

4.4 Module 4: Organisation Management

The Organisation Management module shall provide comprehensive administrative and organisational tools for managing business operations, company structure, scheduling, documentation, and internal communication within each tenant organisation.

Sub-Module 4A: Admin Page / Business Central Portal

Feature ID	Feature Name	Description	Priority
ORG-A-001	Subscription Management	Manage software payment plans and billing cycles	Critical
ORG-A-002	Storage Usage Monitor	Track and manage cloud storage usage per tenant	High
ORG-A-003	User Licence Management	Add/remove user licences and seat management	Critical
ORG-A-004	Database Manager	Admin-level database access and password manager	Critical
ORG-A-005	System Settings	Global configurations and feature toggles	High

Sub-Module 4B: Organisation Structure

Feature ID	Feature Name	Description	Priority
ORG-B-001	Hierarchy Definition	Define departments, teams, and reporting structure	Critical

Feature ID	Feature Name	Description	Priority
ORG-B-002	Role Management	Create custom roles with granular permissions	Critical
ORG-B-003	Organisation Credentials	Store company documents: PAN, GST, Incorporation certificate	High
ORG-B-004	Approval Workflows	Define multi-level approval processes for various transactions	High

Sub-Module 4C: Notice Board

Feature ID	Feature Name	Description	Priority
ORG-C-001	Announcements	Post company-wide or team-specific notices	High
ORG-C-002	Push Notifications	Send notifications for important announcements	High
ORG-C-003	Notice Archive	Historical notices searchable by date/keyword	Medium

Sub-Module 4D: Calendar & Scheduling

Feature ID	Feature Name	Description	Priority
ORG-D-001	Meeting Scheduler	Create, edit, and cancel meetings	Critical
ORG-D-002	Appointment Booking	Schedule appointments with calendar sync	High
ORG-D-003	Reminders	Auto reminders via email/SMS/push notification	High
ORG-D-004	Calendar Integration	Sync with Google Calendar and Outlook Calendar	High
ORG-D-005	Recurring Events	Support for daily/weekly/monthly recurring meetings	Medium

Sub-Module 4E: Notes & To-Do

Feature ID	Feature Name	Description	Priority
ORG-E-001	Personal Notes	Create and manage personal notes with rich text	Medium
ORG-E-002	To-Do Lists	Create task lists with checkboxes and due dates	Medium
ORG-E-003	Shared Notes	Share notes with team members	Low

Sub-Module 4F: Digital Signature

Feature ID	Feature Name	Description	Priority
ORG-F-001	E-Signature	Draw or upload digital signatures	High
ORG-F-002	Signature Requests	Send documents for signature to multiple parties	High
ORG-F-003	Signature Audit Trail	Track signature status with timestamps	High

Sub-Module 4G: Contracts Management

Feature ID	Feature Name	Description	Priority
ORG-G-001	Contract Creation	Create contracts with templates and custom clauses	High
ORG-G-002	Contract Tracking	Track contract status: Draft, Sent, Signed, Expired	High
ORG-G-003	Renewal Alerts	Auto-alerts for upcoming contract renewals	Medium
ORG-G-004	Contract Repository	Centralised storage and search for all contracts	Medium

Sub-Module 4H: Form Builder

Feature ID	Feature Name	Description	Priority
ORG-H-001	Drag-and-Drop Builder	Build custom forms with various field types	Medium
ORG-H-002	Conditional Logic	Show/hide fields based on user input	Medium
ORG-H-003	Form Submission Tracking	View and export form responses	Medium
ORG-H-004	Form Embedding	Embed forms on websites or share via link	Low

Sub-Module 4I: Library / Knowledge Base

Feature ID	Feature Name	Description	Priority
ORG-I-001	Document Library	Upload and categorise documents, SOPs, and policies	High
ORG-I-002	Search & Filter	Full-text search across documents	High
ORG-I-003	Version Management	Track document versions and changes	Medium
ORG-I-004	Access Control	Control who can view/edit documents	High

Sub-Module 4J: Reports Generator

Feature ID	Feature Name	Description	Priority
ORG-J-001	Custom Report Builder	Build reports by selecting data sources, filters, and formats	High
ORG-J-002	Report Templates	Pre-built and reusable report templates	High
ORG-J-003	Scheduled Reports	Auto-generate and email reports on schedule	Medium
ORG-J-004	Export Options	Export reports to PDF, Excel, and CSV	High
ORG-J-005	Dashboard Integration	Pin report widgets to dashboards	Medium

4.5 Module 5: Sales & CRM

The Sales & CRM module shall manage the complete sales lifecycle from lead acquisition through to order fulfilment, invoicing, and recurring subscriptions. It shall also provide point-of-sale capabilities and sales route planning for field teams.

Sub-Module 5A: CRM – Lead & Opportunity Management

Feature ID	Feature Name	Description	Priority
SALES-A-001	Lead Capture	Capture leads from web forms, email, phone, and social media	Critical
SALES-A-002	Lead Pipeline	Visual Kanban pipeline: New, Qualified, Proposal, Negotiation, Won/Lost	Critical
SALES-A-003	Lead Scoring	Auto-score leads based on configurable criteria	High
SALES-A-004	Contact Management	Store contact details, company info, communication history	Critical
SALES-A-005	Activity Tracking	Log calls, emails, meetings, and notes per lead/contact	High
SALES-A-006	Deal/Opportunity Management	Track deal value, probability, expected close date	Critical
SALES-A-007	Sales Forecasting	Predict revenue based on pipeline data	Medium
SALES-A-008	Bulk Import/Export	Import/export leads and contacts via CSV/Excel	High

Sub-Module 5B: Quotations & Invoicing

Feature ID	Feature Name	Description	Priority
SALES-B-001	Quotation Generator	Create professional quotations with templates	Critical
SALES-B-002	Invoice Generation	Generate GST-compliant invoices	Critical

Feature ID	Feature Name	Description	Priority
SALES-B-003	Invoice Templates	Customisable invoice templates with branding	High
SALES-B-004	Payment Terms	Configure payment terms and due dates	High
SALES-B-005	Quotation to Invoice	Convert accepted quotations to invoices	High
SALES-B-006	Multi-Currency Support	Support for international invoicing	Low

Sub-Module 5C: Point of Sale (POS)

Feature ID	Feature Name	Description	Priority
SALES-C-001	POS Interface	Touch-optimised billing interface for retail/restaurants	High
SALES-C-002	Third-Party Order Integration	Integration with Zomato, Swiggy, and food aggregators	Medium
SALES-C-003	Captain Features	Waiter/server order management for restaurants	Medium
SALES-C-004	Table Reservation	Manage table bookings and availability	Medium
SALES-C-005	Token/Point Management	Token system for queues and loyalty points	Low
SALES-C-006	Tally Integration	Export sales data to Tally ERP for accounting	Medium
SALES-C-007	QR Code Functions	Menu display, order placement, feedback, and payment via QR	Medium
SALES-C-008	Kiosk Software	Self-service menu portal for customers	Low
SALES-C-009	Waiter Calling System	Digital waiter call integration	Low

Sub-Module 5D: Subscriptions

Feature ID	Feature Name	Description	Priority
SALES-D-001	Recurring Invoices	Auto-generate invoices for subscriptions	High
SALES-D-002	Renewal Reminders	Alert customers before renewal due date	High
SALES-D-003	Subscription Plans	Define plans: monthly, quarterly, and annual	High

Sub-Module 5E: Route Planning

Feature ID	Feature Name	Description	Priority
SALES-E-001	Sales Route Planner	Plan optimal visit routes for field sales team	Medium
SALES-E-002	Customer Location Map	Map showing customer locations for route planning	Medium
SALES-E-003	Visit Logging	Log sales visits with notes and outcomes	Medium

4.6 Module 6: Marketing Automation

The Marketing Automation module shall provide tools for executing multi-channel marketing campaigns, managing social media presence, organising events, and conducting surveys to support lead generation and customer engagement activities.

Sub-Module 6A: Email & SMS Marketing

Feature ID	Feature Name	Description	Priority
MKTG-A-001	Email Campaign Builder	Drag-and-drop email designer with visual editor	High
MKTG-A-002	Email Sending	Send bulk emails to segmented contact lists	High
MKTG-A-003	Email Tracking	Track open rates, click rates, and bounces	High
MKTG-A-004	SMS Campaign Builder	Create and schedule SMS campaigns	High
MKTG-A-005	SMS Tracking	Delivery status and response tracking	High
MKTG-A-006	WhatsApp Marketing	Send campaigns via WhatsApp Business API	Medium
MKTG-A-007	Contact Segmentation	Segment contacts by tags, behaviour, and demographics	High
MKTG-A-008	A/B Testing	Test subject lines and content variations	Medium

Sub-Module 6B: Social Media Management

Feature ID	Feature Name	Description	Priority
MKTG-B-001	Social Media Posting	Schedule posts to Facebook, LinkedIn, and Twitter (API costs applicable)	Medium
MKTG-B-002	Website Visitor Tracking	Track visitor behaviour and heatmaps	Medium
MKTG-B-003	Social Listening	Monitor brand mentions and engagement	Low

Sub-Module 6C: Events Management

Feature ID	Feature Name	Description	Priority
MKTG-C-001	Event Creation	Create and publish events	Medium
MKTG-C-002	Ticket Sales	Sell tickets online with payment integration	Medium
MKTG-C-003	Attendee Management	Track registrations and check-ins	Medium

Sub-Module 6D: Surveys

Feature ID	Feature Name	Description	Priority
MKTG-D-001	Survey Builder	Create multi-question surveys	Medium
MKTG-D-002	Survey Distribution	Share via email, link, or embed on website	Medium
MKTG-D-003	Response Analytics	Analyse survey responses with charts	Medium

4.7 Module 7: Supply Chain Management (SCM)

The Supply Chain Management module shall provide end-to-end management of inventory, manufacturing processes, product lifecycle, equipment maintenance, and quality control operations.

Sub-Module 7A: Inventory Management

Feature ID	Feature Name	Description	Priority
SCM-A-001	Stock Management	Track inventory levels, stock in/out	Critical
SCM-A-002	Multi-Location Inventory	Manage inventory across multiple warehouses	High
SCM-A-003	SKU/Product Management	Product master data with variants	Critical
SCM-A-004	Low Stock Alerts	Auto-alerts when stock falls below threshold	High
SCM-A-005	Inventory Valuation	FIFO, LIFO, and Weighted Average methods	High
SCM-A-006	Barcode/QR Scanning	Scan products for faster stock updates	Medium

Sub-Module 7B: Manufacturing

Feature ID	Feature Name	Description	Priority
SCM-B-001	Manufacturing Orders	Create and track production orders	High
SCM-B-002	Bill of Materials (BOM)	Define BOM for manufactured products	High
SCM-B-003	Work Order Scheduling	Schedule production jobs	Medium
SCM-B-004	Production Tracking	Track work-in-progress and completion status	Medium

Sub-Module 7C: Product Lifecycle Management (PLM)

Feature ID	Feature Name	Description	Priority
SCM-C-001	Product Design Management	Manage product designs, specifications, and drawings	Medium
SCM-C-002	Change Management	Track design changes and approvals	Medium
SCM-C-003	Version Control	Maintain product version history	Low

Sub-Module 7D: Maintenance

Feature ID	Feature Name	Description	Priority
SCM-D-001	Asset/Equipment Tracking	Track machinery, vehicles, and equipment	High
SCM-D-002	Maintenance Requests	Submit and track maintenance tickets	High
SCM-D-003	Preventive Maintenance	Schedule regular maintenance based on usage/time	Medium
SCM-D-004	Spare Parts Management	Track spare parts inventory	Medium

Sub-Module 7E: Quality Control

Feature ID	Feature Name	Description	Priority
SCM-E-001	Quality Checks	Define quality control checkpoints	Medium
SCM-E-002	Defect Tracking	Log defects and corrective actions	Medium
SCM-E-003	Quality Reports	Generate quality compliance reports	Low

4.8 Module 8: Human Resource Management (HRM)

The Human Resource Management module shall provide comprehensive tools for managing the entire employee lifecycle, from recruitment and onboarding through attendance tracking, leave management, performance appraisals, and fleet management.

Sub-Module 8A: Employee Management

Feature ID	Feature Name	Description	Priority
HRM-A-001	Employee Database	Centralised employee records with personal/professional information	Critical
HRM-A-002	Employee Onboarding	Digital onboarding workflows with document collection	High
HRM-A-003	Organisation Chart	Visual organisational chart showing reporting structure	Medium
HRM-A-004	Employee Self-Service	Employees update personal info and view payslips	High

Sub-Module 8B: Recruitment

Feature ID	Feature Name	Description	Priority
HRM-B-001	Job Posting	Post vacancies on careers page and job portals	High
HRM-B-002	Applicant Tracking	Track candidates through recruitment pipeline	High
HRM-B-003	Interview Scheduling	Schedule interviews with calendar sync	Medium
HRM-B-004	Candidate Evaluation	Score and compare candidates	Medium

Sub-Module 8C: Holidays & Time Off

Feature ID	Feature Name	Description	Priority
HRM-C-001	Holiday Calendar	Define public holidays and company holidays	High
HRM-C-002	Leave Types	Configure leave types: Casual, Sick, Earned, LOP	Critical
HRM-C-003	Leave Requests	Employees apply for leave via self-service portal	Critical
HRM-C-004	Leave Approval Workflow	Multi-level approval for leave requests	High
HRM-C-005	Leave Balance Tracking	Real-time leave balance for employees	Critical

Sub-Module 8D: Attendance & Time Tracking

Feature ID	Feature Name	Description	Priority
HRM-D-001	Clock In/Out	Mobile and web-based attendance marking	Critical
HRM-D-002	Biometric Integration	Integration with biometric attendance devices	Low
HRM-D-003	GPS Attendance	Location-based attendance for field employees	Medium
HRM-D-004	Attendance Reports	Monthly attendance reports per employee/department	High
HRM-D-005	Overtime Tracking	Track and approve overtime hours	Medium

Sub-Module 8E: Appraisals & Performance

Feature ID	Feature Name	Description	Priority
HRM-E-001	Performance Reviews	Conduct periodic performance appraisals	High
HRM-E-002	Goal Setting	Set SMART goals for employees	Medium
HRM-E-003	360-Degree Feedback	Multi-rater feedback system	Low
HRM-E-004	Performance Reports	Generate performance summary reports	Medium

Sub-Module 8F: Fleet Management

Feature ID	Feature Name	Description	Priority
HRM-F-001	Vehicle Database	Manage company vehicles with details	Medium
HRM-F-002	Fuel & Running Costs	Track fuel consumption and maintenance costs	Medium
HRM-F-003	Vehicle Assignment	Assign vehicles to employees/projects	Medium
HRM-F-004	GPS Tracking	Real-time vehicle location tracking	Low

4.9 Module 9: Finance & Accounting

The Finance & Accounting module shall provide comprehensive financial management capabilities including double-entry accounting, invoicing, payment processing, expense management, payroll with Indian statutory compliance, bill settlements, and financial document management.

Sub-Module 9A: Accounting Books

Feature ID	Feature Name	Description	Priority
FIN-A-001	Chart of Accounts	Define GL accounts with hierarchy	Critical
FIN-A-002	Journal Entries	Manual and auto journal entry recording	Critical
FIN-A-003	General Ledger	View GL with drill-down to transactions	Critical
FIN-A-004	Trial Balance	Generate trial balance reports	Critical
FIN-A-005	Financial Statements	Auto-generate Profit & Loss, Balance Sheet, and Cash Flow statements	Critical
FIN-A-006	Multi-Currency Support	Handle foreign currency transactions	Low
FIN-A-007	Bank Reconciliation	Match bank statements with book entries	High

Sub-Module 9B: Invoicing & Payments

Feature ID	Feature Name	Description	Priority
FIN-B-001	Sales Invoices	Generate customer invoices (GST-compliant)	Critical
FIN-B-002	Purchase Invoices	Record supplier invoices	Critical
FIN-B-003	Payment Recording	Record customer payments against invoices	Critical
FIN-B-004	Payment Gateway Integration	Accept online payments via Razorpay/Stripe	High
FIN-B-005	Credit/Debit Notes	Issue credit/debit notes for adjustments	High
FIN-B-006	Aging Reports	Track receivables and payables aging	High

Sub-Module 9C: Expense Management

Feature ID	Feature Name	Description	Priority
FIN-C-001	Expense Submission	Employees submit expense claims with receipts	High
FIN-C-002	Expense Approval	Multi-level approval workflow for expenses	High
FIN-C-003	Expense Reimbursement	Track reimbursement status and payments	High
FIN-C-004	Expense Categories	Configure expense categories and limits	Medium

Sub-Module 9D: Payroll

Feature ID	Feature Name	Description	Priority
FIN-D-001	Salary Structures	Define salary components: Basic, HRA, Allowances, and Deductions	Critical
FIN-D-002	Payroll Processing	Auto-calculate monthly payroll with attendance integration	Critical
FIN-D-003	Payslip Generation	Generate and email payslips to employees	Critical
FIN-D-004	Statutory Compliance	Calculate PF, ESI, TDS, and Professional Tax	Critical
FIN-D-005	Payroll Reports	Generate payroll registers and tax reports	High
FIN-D-006	Bank Transfer File	Generate bank transfer files for salary disbursement	High

Sub-Module 9E: Bill Settlements

Feature ID	Feature Name	Description	Priority
FIN-E-001	Vendor Bill Management	Record and track vendor bills	High
FIN-E-002	Payment Scheduling	Schedule bill payments	High
FIN-E-003	Payment Approval	Approval workflow for payments	High

Sub-Module 9F: Document Management

Feature ID	Feature Name	Description	Priority
FIN-F-001	Financial Document Storage	Store invoices, receipts, and financial documents	High
FIN-F-002	Document Categorisation	Categorise and tag financial documents	Medium
FIN-F-003	Audit Trail	Complete audit trail for all financial transactions	Critical

4.10 Module 10: Task/Services/Project Management

The Task/Services/Project Management module shall provide comprehensive project planning, task tracking, timesheet management, field service management, helpdesk/ticketing, employee scheduling, and project document management capabilities.

Sub-Module 10A: Project Management

Feature ID	Feature Name	Description	Priority
PM-A-001	Project Creation	Create projects with scope, timeline, and budget	Critical
PM-A-002	Task Management	Create tasks with assignments, due dates, and priorities	Critical
PM-A-003	Gantt Charts	Visual timeline with dependencies	High
PM-A-004	Kanban Boards	Drag-and-drop task boards (To Do, In Progress, Done)	High
PM-A-005	Milestone Tracking	Define and track project milestones	High
PM-A-006	Project Templates	Save and reuse project templates	Medium
PM-A-007	Resource Allocation	Assign team members and resources to tasks	High
PM-A-008	Project Dashboard	Real-time project status overview	High

Sub-Module 10B: Timesheet Management

Feature ID	Feature Name	Description	Priority
PM-B-001	Time Logging	Employees log hours per task/project	High
PM-B-002	Timesheet Approval	Manager approves submitted timesheets	High
PM-B-003	Billable vs Non-Billable	Categorise time as billable or internal	Medium
PM-B-004	Time Reports	Generate utilisation and productivity reports	High

Sub-Module 10C: Field Service Management

Feature ID	Feature Name	Description	Priority
PM-C-001	Work Order Management	Create and dispatch field work orders	High
PM-C-002	Technician Assignment	Assign jobs to field technicians	High
PM-C-003	Time & Material Tracking	Track time and materials used on-site	Medium
PM-C-004	Mobile App for Field Team	Mobile interface for field updates	High
PM-C-005	Customer Signature Capture	Capture customer signature on job completion	Medium

Sub-Module 10D: Helpdesk / Ticketing

Feature ID	Feature Name	Description	Priority
PM-D-001	Ticket Creation	Customers/employees raise support tickets	Critical
PM-D-002	Ticket Assignment	Auto-assign or manually assign tickets to agents	Critical
PM-D-003	Ticket Priority & SLA	Prioritise tickets and track SLA compliance	High
PM-D-004	Ticket Status Tracking	Track ticket lifecycle: Open, In Progress, Resolved	Critical
PM-D-005	Knowledge Base Integration	Link solutions from knowledge base	Medium
PM-D-006	Customer Portal	Customer self-service portal to view tickets	Medium

Sub-Module 10E: Planning / Scheduling

Feature ID	Feature Name	Description	Priority
PM-E-001	Employee Schedule	Create weekly/monthly work schedules	Medium
PM-E-002	Shift Management	Define shifts and assign employees	Medium
PM-E-003	Schedule Conflicts	Alert on scheduling conflicts	Low

Sub-Module 10F: Project Database / File Management

Feature ID	Feature Name	Description	Priority
PM-F-001	Project File Storage	Store project-related documents, drawings, and reports	High
PM-F-002	Version Control	Track document versions	Medium
PM-F-003	File Sharing	Share files with team members and clients	High
PM-F-004	Search & Filter	Search files by name, tag, and date	Medium

4.11 Module 11: Website Builder & Content Management

The Website Builder & Content Management module shall enable tenant organisations to create and manage their web presence, publish blog content, host community forums, and engage visitors through live chat, all within the TixelERP platform.

Sub-Module 11A: Website Builder

Feature ID	Feature Name	Description	Priority
WEB-A-001	Drag-and-Drop Builder	Visual website builder with WYSIWYG editor	Medium

Feature ID	Feature Name	Description	Priority
WEB-A-002	Static & Dynamic Pages	Support for static content and dynamic data	Medium
WEB-A-003	Template Library	Pre-built website templates	Medium
WEB-A-004	Mobile Responsive	Auto-responsive design for mobile devices	High
WEB-A-005	SEO Tools	Meta tags, sitemaps, and SEO optimisation	Medium

Sub-Module 11B: Blog Management

Feature ID	Feature Name	Description	Priority
WEB-B-001	Blog Post Creation	Rich text editor for blog posts	Medium
WEB-B-002	Categories & Tags	Organise posts with categories and tags	Medium
WEB-B-003	Comments Management	Moderate and respond to comments	Low
WEB-B-004	RSS Feed	Auto-generate RSS feeds	Low

Sub-Module 11C: Forum / Q&A

Feature ID	Feature Name	Description	Priority
WEB-C-001	Discussion Forum	Create forum topics and threads	Low
WEB-C-002	FAQ Management	Create and manage FAQs	Medium
WEB-C-003	Upvoting & Best Answers	Community-driven Q&A features	Low

Sub-Module 11D: Live Chat

Feature ID	Feature Name	Description	Priority
WEB-D-001	Live Chat Widget	Embeddable chat widget for website	Medium
WEB-D-002	Chat Routing	Route chats to available agents	Medium
WEB-D-003	Chat History	Store and review chat transcripts	Medium
WEB-D-004	Canned Responses	Quick reply templates for common queries	Low

4.12 Module 12: Report Generator

The Report Generator module shall provide industry-specific and custom report generation capabilities, with an initial focus on civil industry reports, extensible to other industries as required.

Feature ID	Feature Name	Description	Priority
RPT-A-001	Survey Reports	Generate land survey reports	Low
RPT-A-002	Geotechnical Reports	Soil testing and foundation reports	Low
RPT-A-003	Design Reports	Structural design calculation reports	Low
RPT-A-004	Custom Report Templates	Industry-specific configurable templates	Low

4.13 Module 13: Office / Workspace Collaboration Tools

The Office / Workspace Collaboration Tools module shall provide an integrated suite of productivity and communication tools enabling employees to create documents, collaborate in real time, communicate via messaging and calls, and conduct video conferences within the TixelERP platform.

Sub-Module 13A: Word Processor

Feature ID	Feature Name	Description	Priority
OFFICE-A-001	Rich Text Editor	MS Word-like editor with formatting	Medium
OFFICE-A-002	Document Templates	Pre-built templates for letters and reports	Medium
OFFICE-A-003	Collaboration	Real-time collaborative editing	Low
OFFICE-A-004	Export to DOCX/PDF	Export documents to standard formats	High

Sub-Module 13B: Spreadsheet

Feature ID	Feature Name	Description	Priority
OFFICE-B-001	Excel-like Grid	Spreadsheet with formulas and formatting	Medium
OFFICE-B-002	Charts & Pivot Tables	Data visualisation within spreadsheet	Low
OFFICE-B-003	Export to XLSX	Export to Excel format	High

Sub-Module 13C: Presentation

Feature ID	Feature Name	Description	Priority
OFFICE-C-001	Slide Editor	PowerPoint-like presentation creator	Low
OFFICE-C-002	Themes & Layouts	Pre-designed presentation themes	Low
OFFICE-C-003	Export to PPTX/PDF	Export presentations to standard formats	Medium

Sub-Module 13D: Email Platform

Feature ID	Feature Name	Description	Priority
OFFICE-D-001	Email Client	Basic email client (or integrate with Gmail/Outlook)	Low
OFFICE-D-002	Email Templates	Reusable email templates	Medium

Sub-Module 13E: Internal Messaging (Discuss)

Feature ID	Feature Name	Description	Priority
OFFICE-E-001	Group Channels	Create public/private channels for teams	High
OFFICE-E-002	Direct Messaging	1-on-1 messaging between employees	High
OFFICE-E-003	File Sharing	Share files within chat	High
OFFICE-E-004	Search History	Search past messages	Medium
OFFICE-E-005	Mentions & Notifications	@mentions with push notifications	High

Sub-Module 13F: Voice Calls

Feature ID	Feature Name	Description	Priority
OFFICE-F-001	VoIP Calling	Make and receive voice calls within app	Low
OFFICE-F-002	Call Recording	Record calls for training/quality (with consent)	Low

Sub-Module 13G: Video Conferencing

Feature ID	Feature Name	Description	Priority
OFFICE-G-001	Video Meetings	Host video conferences (configurable participant limit)	Medium
OFFICE-G-002	Screen Sharing	Share screen during calls	Medium
OFFICE-G-003	Meeting Recording	Record meetings	Low
OFFICE-G-004	Chat During Calls	Text chat alongside video	Medium

5. CROSS-CUTTING PLATFORM REQUIREMENTS

The following requirements apply across all modules and represent foundational platform capabilities.

5.1 Core Platform Features

Feature ID	Feature Name	Description	Priority
CORE-001	Multi-Tenant Architecture	Support for multiple organisations on a single platform with logical data isolation	Critical
CORE-002	Mobile Apps (iOS & Android)	Flutter or hybrid mobile applications with feature parity where technically feasible	Critical
CORE-003	Offline Mode	Basic offline functionality with automatic synchronisation upon reconnection	Medium
CORE-004	Advanced Search	Global search across all modules with filters and faceted results	High
CORE-005	Export/Import	Bulk data export/import in Excel and CSV formats	High
CORE-006	Activity Logs	System-wide audit trail recording all significant user and system actions	Critical
CORE-007	Notifications Engine	Unified notification system: email, SMS, push, and in-app	Critical
CORE-008	API Access	RESTful APIs for third-party integrations with authentication and rate limiting	High
CORE-009	Webhooks	Event-driven webhooks for automation and integration	Medium
CORE-010	Custom Fields	Allow custom fields across modules for tenant-specific data requirements	Medium

5.2 Security & Compliance Features

Feature ID	Feature Name	Description	Priority
SEC-001	Data Encryption	AES-256 encryption at rest; TLS 1.2+ in transit	Critical
SEC-002	RBAC (Role-Based Access)	Granular permissions per module/feature	Critical
SEC-003	IP Whitelisting	Restrict access by IP address	Medium
SEC-004	Session Timeout	Auto-logout after configurable period of inactivity	High
SEC-005	DPDP Compliance	Compliance with Digital Personal Data Protection Act, 2023 (India)	Critical
SEC-006	GDPR Readiness	GDPR compliance for international clients	Low
SEC-007	Data Retention Policies	Configure data retention and purging policies	Medium

5.3 Performance & Scalability Features

Feature ID	Feature Name	Description	Priority
PERF-001	Caching	Redis/Memcached for performance optimisation	Critical
PERF-002	Load Balancing	Distribute load across servers	High
PERF-003	CDN Integration	Content delivery via CDN for faster loading	Medium
PERF-004	Database Optimisation	Indexing, query optimisation, and connection pooling	Critical

6. EXTERNAL INTERFACE REQUIREMENTS

6.1 User Interfaces

The system shall provide a web-based graphical user interface designed to be intuitive, responsive, and accessible. The user interface shall comprise the following key components: a secure login page with MFA support; a personalised dashboard with configurable widgets; a structured navigation menu providing access to all modules; data entry forms with client-side validation and real-time feedback; search and filtering tools across all modules; reporting and visualisation interfaces with interactive charts; and a responsive design supporting desktops, laptops, tablets, and mobile devices.

All user interfaces shall comply with WCAG 2.1 Level AA accessibility standards and shall support keyboard navigation, screen reader compatibility, and colour contrast requirements.

6.2 Hardware Interfaces

As a web-based SaaS platform, TixelERP shall have minimal hardware interface requirements. Users shall access the system through standard computing devices (desktops, laptops, tablets, and mobile phones) with modern web browsers. The application server and database server infrastructure shall be provisioned on cloud platforms. Integration with barcode/QR code scanners (for inventory management) and biometric devices (for attendance tracking) shall be supported through standard communication protocols.

6.3 Software Interfaces

TixelERP shall interface with the following categories of external software systems:

Interface Category	Examples	Protocol
Database Management System	MongoDB	Native MongoDB connector
Payment Gateways	Razorpay, Stripe, PayU (as approved by TixelTech)	REST API / Webhooks
Email Service Providers	SendGrid, AWS SES (or similar)	SMTP / REST API
SMS Gateways	Twilio, MSG91 (or similar)	REST API
WhatsApp Business API	Official WhatsApp Business API	REST API
Cloud Storage	AWS S3, Google Cloud Storage (or similar)	SDK / REST API
Calendar Services	Google Calendar, Outlook Calendar	CalDAV / REST API
Social Media Platforms	Facebook, LinkedIn, Twitter	Platform-specific APIs
Accounting Software	Tally ERP	Data export / API integration
Food Aggregators	Zomato, Swiggy (Optional)	REST API
Map/Geolocation Services	Google Maps API (or equivalent)	REST API / SDK
Web Browsers	Chrome, Firefox, Safari, Edge (latest 2 versions)	HTTP/HTTPS

6.4 Communication Interfaces

All client-server communication shall occur over HTTPS (TLS 1.2 or higher). The system shall expose RESTful APIs following JSON data format standards for all inter-service and client-server communication. Real-time features (notifications, chat, live tracking) shall utilise WebSocket connections or Server-Sent Events (SSE) as appropriate. Email and SMS communications shall utilise the respective service provider APIs. Push notifications shall be delivered through platform-specific push notification services (Firebase Cloud Messaging for Android, Apple Push Notification Service for iOS).

7. NON-FUNCTIONAL REQUIREMENTS

7.1 Performance Requirements

The system shall meet the following performance benchmarks, as stipulated in Clause 4.5.1(d) of the Governing Agreement:

Parameter	Requirement
Page Load Time	Less than 3 seconds for standard operations under normal load conditions
API Response Time	Less than 500 milliseconds for standard queries
Concurrent Users	The system shall support a minimum of 500 concurrent users per tenant without degradation
Data Processing	Transactions such as sales orders, inventory updates, and financial records shall be processed without perceptible delay
Report Generation	Standard reports shall generate within 10 seconds; complex reports within 30 seconds
Database Query Optimisation	All frequent queries shall be optimised with appropriate indexing and caching
Resource Utilisation	The application shall be optimised for efficient use of server resources (CPU, memory, database connections)

7.2 Security Requirements

The system shall implement the following security measures in compliance with OWASP guidelines and the security standards specified in the Governing Agreement:

- (a) User authentication shall be enforced via OAuth 2.0 / JWT with optional multi-factor authentication (TOTP per RFC 6238).
- (b) Role-Based Access Control (RBAC) shall restrict user access to modules, features, and data elements based on assigned roles.
- (c) All data in transit shall be encrypted using TLS 1.2 or higher; all data at rest shall be encrypted using AES-256.
- (d) Strong password policies shall be enforced, including configurable minimum length, complexity, and expiry requirements.
- (e) Account lockout shall be triggered after a configurable number of failed login attempts, with CAPTCHA support.
- (f) Comprehensive audit logging shall record all significant user activities including login/logout, data modifications, and system configuration changes.
- (g) Input validation and sanitisation shall protect against OWASP Top 10 vulnerabilities including MongoDB injection, Cross-Site Scripting (XSS), Cross-Site Request Forgery (CSRF), and others.
- (h) The system shall comply with the Digital Personal Data Protection Act, 2023 (India) and shall be designed with GDPR readiness for international clients.

7.3 Reliability Requirements

- (a) The system shall operate continuously without frequent crashes or unexpected shutdowns under normal operational conditions.
- (b) Error handling shall be implemented to detect errors gracefully, display meaningful error messages to users, and prevent data corruption.
- (c) Data integrity shall be maintained during all operations through ACID-compliant database transactions.
- (d) Automated daily backups shall be performed, with the ability to restore data within a defined Recovery Point Objective (RPO) and Recovery Time Objective (RTO).
- (e) Transaction reliability shall ensure that financial operations and order processing are completed fully and accurately without partial updates.

7.4 Scalability Requirements

- (a) The system shall support horizontal scaling to accommodate increasing numbers of tenants, users, and data volumes.
- (b) The modular architecture shall allow additional modules or features to be integrated without affecting existing system functionality.
- (c) The database shall be capable of handling growth in records to millions of entries without significant performance degradation.
- (d) Infrastructure scaling shall be supported through the cloud hosting platform (auto-scaling, additional compute/storage resources).

7.5 Availability Requirements

- (a) The system shall maintain a target uptime of 99.5% measured on a monthly basis, excluding scheduled maintenance windows.
- (b) Planned maintenance shall be scheduled during off-peak hours (typically 12:00 AM to 6:00 AM IST) with at least 48 hours advance notice to affected tenants.
- (c) The infrastructure shall include redundancy and failover mechanisms to minimise unplanned downtime.
- (d) System monitoring shall be implemented to detect and alert on issues proactively.

7.6 Usability Requirements

- (a) The system shall provide a clean, intuitive graphical user interface enabling users to navigate efficiently across modules.
- (b) New users shall be able to learn and use basic system functions with minimal training.
- (c) The interface shall maintain consistent layout, navigation, and design elements across all modules.
- (d) Clear and actionable error messages shall be displayed for invalid user actions or system errors.

- (e) Help sections, tooltips, and contextual guidance shall be provided to assist users.
- (f) The interface shall be accessible to users with varying levels of technical expertise and shall comply with WCAG 2.1 Level AA standards.

7.7 Maintainability Requirements

- (a) The codebase shall follow clean coding practices with meaningful variable, function, and class names, adequate inline documentation, and adherence to industry-standard style guides.
- (b) Version control shall be maintained using Git with a GitFlow branching strategy and meaningful commit messages.
- (c) The system architecture shall support modular development, enabling independent development, testing, and deployment of individual modules.
- (d) Comprehensive API documentation shall be maintained using Swagger/OpenAPI specification.

8. DATABASE DESIGN CONSIDERATIONS

The TixelERP database shall be designed on MongoDB with the following considerations:

- (a) A shared-database, shared-schema multi-tenant design shall be implemented, with a tenant identifier column on all tenant-scoped tables.
- (b) Application-level Row-Level Security (RLS) shall be enforced through a scoped Express QuerySet manager to prevent cross-tenant data access.
- (c) The database schema shall be normalised to at least Third Normal Form (3NF) for core transactional tables, with selective denormalisation permitted for reporting and analytics tables where performance justifies it.
- (d) Appropriate indexing strategies shall be implemented on frequently queried columns, foreign keys, and tenant identifier columns.
- (e) Database migrations shall be managed through Express's built-in migration framework, ensuring version-controlled and reversible schema changes.
- (f) Entity-Relationship (ER) diagrams and complete database schema documentation shall be maintained as part of the technical documentation deliverables.
- (g) Sensitive data fields (passwords, financial information, personal data) shall be encrypted at the application layer before storage.

9. DEPLOYMENT REQUIREMENTS

9.1 Environment Configuration

Three separate environments shall be provisioned and maintained throughout the project lifecycle:

Environment	Purpose	Access
Development	Active development and unit testing by Trackepay development team	Trackepay team only
Staging	Integration testing, UAT, and pre-production validation	Trackepay and TixelTech teams
Production	Live production environment serving end users	Controlled access with change management

9.2 CI/CD Pipeline

A Continuous Integration / Continuous Deployment (CI/CD) pipeline shall be established to automate code building, testing, and deployment. The pipeline shall include automated unit test execution, code quality and linting checks, security vulnerability scanning, automated deployment to staging, and manual approval gates for production deployments.

9.3 Infrastructure Requirements

The production infrastructure shall include: application servers with auto-scaling capability; managed MongoDB database with automated backups; Redis cache cluster; object storage

for file/document management; CDN for static asset delivery; SSL/TLS certificates; load balancer for traffic distribution; monitoring, logging, and alerting systems; and automated backup with configurable retention policies.

10. ACCEPTANCE CRITERIA & QUALITY ASSURANCE

In accordance with Clause 4.8 and Annexure A (Section A.16) of the Governing Agreement, each module, feature, and deliverable shall be considered complete and acceptable only when the following criteria are satisfied:

- (a) **Functionality:** The deliverable matches the specifications set forth in Section 4 of this SRS and Annexure A of the Governing Agreement.
- (b) **Quality:** No Critical (P1) or High (P2) severity bugs remain unresolved.
- (c) **Performance:** The deliverable meets the performance standards specified in Section 7.1 (page load < 3 seconds, API response < 500 milliseconds).
- (d) **Security:** The deliverable passes security review and vulnerability scanning in accordance with Section 7.2.
- (e) **Documentation:** User and technical documentation for the deliverable is complete and accurate.
- (f) **UAT Approval:** The deliverable passes User Acceptance Testing conducted by TixelTech.

10.1 Testing Strategy

Testing Type	Description	Responsibility
Unit Testing	Testing of individual functions, methods, and components	Trackepay
Integration Testing	Testing inter-module communication and data flow	Trackepay
Functional Testing	Testing all user-facing features against requirements	Trackepay
Regression Testing	Re-testing after changes to ensure no existing functionality is broken	Trackepay
Performance Testing	Load testing and stress testing to validate performance benchmarks	Trackepay
Security Testing	Vulnerability assessments and penetration testing	Trackepay
Browser Compatibility	Cross-browser testing on Chrome, Firefox, Safari, and Edge	Trackepay
Mobile Device Testing	Cross-device testing on Android and iOS devices	Trackepay
User Acceptance Testing	Validation of deliverables against business requirements by TixelTech	TixelTech (with Trackepay support)

11. PHASED DELIVERY PLAN

The development and delivery of TixelERP shall follow a phased approach as defined in Annexure A (Section A.15) and Annexure F of the Governing Agreement:

Phase	Modules	Milestone	Target Completion
Phase 1: MVP	Module 1 (Authentication), Module 2 (Home Screen), Module 3 (Dashboard – core views), Module 4 (Organisation Management – basic), Module 9 (Finance – basic invoicing/accounting)	M2	To Be Determined
Phase 2: Core Business	Module 5 (Sales & CRM), Module 8 (HRM – employees, attendance, leave), Module 10 (Project Management)	M3	To Be Determined
Phase 3: Extended Features	Module 6 (Marketing Automation), Module 7 (SCM), Module 11 (Website Builder), Module 13 (Collaboration Tools)	M4	To Be Determined
Phase 4: Advanced & Industry-Specific	Module 12 (Report Generator – industry-specific), Advanced features across all modules, Performance optimisation, Final UAT and Go-Live preparation	M5	To Be Determined

12. PAYMENT MILESTONES & SRS DELIVERABLE LINKAGE

This SRS document is a designated deliverable under Milestone M1 (Kickoff & Planning) of Annexure F of the Governing Agreement. The complete milestone-based payment schedule, as agreed between the Parties, is reproduced below for reference:

Milestone	Deliverable	% of Total	Amount (₹)	Due Date
M1: Kickoff & Planning	Project kickoff, technology stack approval, architecture document (HLD), SRS document, development environment setup	20%	1,60,000	Within 7 days of acceptance
M2: Phase 1 – MVP Core	Modules 1–4 + basic Finance; Alpha version on staging	20%	1,60,000	Within 7 days of acceptance
M3: Phase 2 – Business Modules	Sales/CRM, HRM, Project Management; Beta version on staging	20%	1,60,000	Within 7 days of acceptance
M4: Phase 3 – Extended Features	Marketing Automation, SCM, Website Builder, Collaboration tools; Integration testing complete	20%	1,60,000	Within 7 days of acceptance
M5: Go-Live	Final UAT passed, production deployment, training completed, all documentation delivered	10%	80,000	Within 7 days of Go-Live
M6: Retention Period	Retention release upon 30 consecutive days with no critical/high defects	10%	80,000	6 months after Go-Live

Total Development Consideration: ₹8,00,000/- (Rupees Eight Lakhs Only)

Upon written acceptance of this SRS and the associated M1 deliverables by TixelTech, Trackepay shall raise a GST invoice for ₹1,60,000/- (Rupees One Lakh Sixty Thousand Only), payable within seven (7) days of the invoice date via electronic bank transfer to Trackepay's designated account.

13. SYSTEM CONSTRAINTS

The ERP system shall operate within the following constraints that shall be considered during system implementation:

Technology Constraints: The system shall be developed using the approved technology stack (Next.js, React, Express Js, Node, MongoDB) with no deviation without prior written approval from TixelTech.

Hardware Constraints: The system shall be compatible with standard computing devices and cloud infrastructure. Specific hardware integrations (barcode scanners, biometric devices) shall operate through standard communication protocols.

Network Constraints: The system requires stable internet connectivity for full functionality. Offline mode shall provide limited functionality with synchronisation upon reconnection.

Security and Compliance Constraints: The system shall comply with OWASP security guidelines, DPDP Act 2023, and GST regulations for invoicing.

Budget Constraints: Development shall remain within the approved budget of ₹8,00,000/- with a cost cap of ₹10,00,000/- inclusive of approved change requests.

Time Constraints: The system development and deployment shall be completed within the agreed project timeline (to be finalised in Annexure F of the Governing Agreement).

14. FUTURE ENHANCEMENTS

The following capabilities have been identified as potential future enhancements beyond the initial scope defined in this SRS. These enhancements shall be subject to mutual agreement, formal Change Request procedures (Annexure G), and separate cost assessment:

- (a) Artificial Intelligence and Machine Learning integration for predictive analytics, intelligent lead scoring, demand forecasting, and anomaly detection.
- (b) Advanced business intelligence with custom dashboard builder, drill-down analytics, and scheduled report automation.
- (c) IoT (Internet of Things) integration for real-time sensor data from manufacturing equipment, fleet tracking, and environmental monitoring.
- (d) Blockchain integration for secure document verification, supply chain traceability, and smart contract execution.
- (e) Advanced workflow automation engine with visual workflow designer and rule-based triggers.
- (f) Multi-language expansion beyond English and Hindi to support regional Indian languages and international languages.

- (g) Industry-specific vertical modules (e.g., healthcare, education, real estate) tailored to sector-specific requirements.
- (h) Advanced mobile features including augmented reality (AR) for field service and inventory management.

15. DOCUMENT APPROVAL & SIGN-OFF

This Software Requirements Specification (SRS) document has been reviewed and is hereby approved by the undersigned authorised representatives of both Parties. By signing below, each Party confirms that the requirements, scope, specifications, and terms described herein are complete, accurate, and agreed upon, and that this document shall serve as the authoritative baseline for the design, development, testing, and acceptance of TixelERP.

FOR AND ON BEHALF OF TIXELTECH PRIVATE LIMITED

Name:	
Designation:	
Date:	
Signature:	

FOR AND ON BEHALF OF TRACKEPAY FINTECH (OPC) PRIVATE LIMITED

Name:	Biswajit Sahoo
Designation:	Project Manager
Date:	
Signature:	

Reviewed & Approved By (Trackepay CTO/Tech Lead):

Name:	Biswajit Pradhan
Designation:	CTO / Tech Lead
Date:	
Signature:	